

The Popcorn Metaphor

A Framework for Understanding Order, Entropy, and Phase Transitions in Granular Systems

Maayan Keynan

Independent Researcher

January 2026

Abstract

This document presents the Popcorn Metaphor, a conceptual framework for understanding order, entropy, and phase transitions in granular systems. Using popcorn kernels as a physical model, we describe how the presence or absence of a coupling medium (oil) fundamentally alters spatial organization, shifting systems from hyperuniform (repulsion-dominated) to clustered (attraction-dominated) states. We propose three testable claims regarding spatial distribution, temporal dynamics of phase transitions, and convergence to jammed states. The framework connects to Voronoi tessellation analysis, offering measurable predictions for entropy scaling. We further describe the multi-scale duality of popped popcorn – foam at the micro-scale, jammed granular matter at the macro-scale – and employ a snow-packing analogy to illustrate the spectrum of jammed configurations.

Keywords: granular systems, hyperuniformity, Voronoi tessellation, phase transitions, jamming, entropy, clustering, self-organization, statistical physics, soft matter physics, capillary bridges, complex systems, packing fraction.

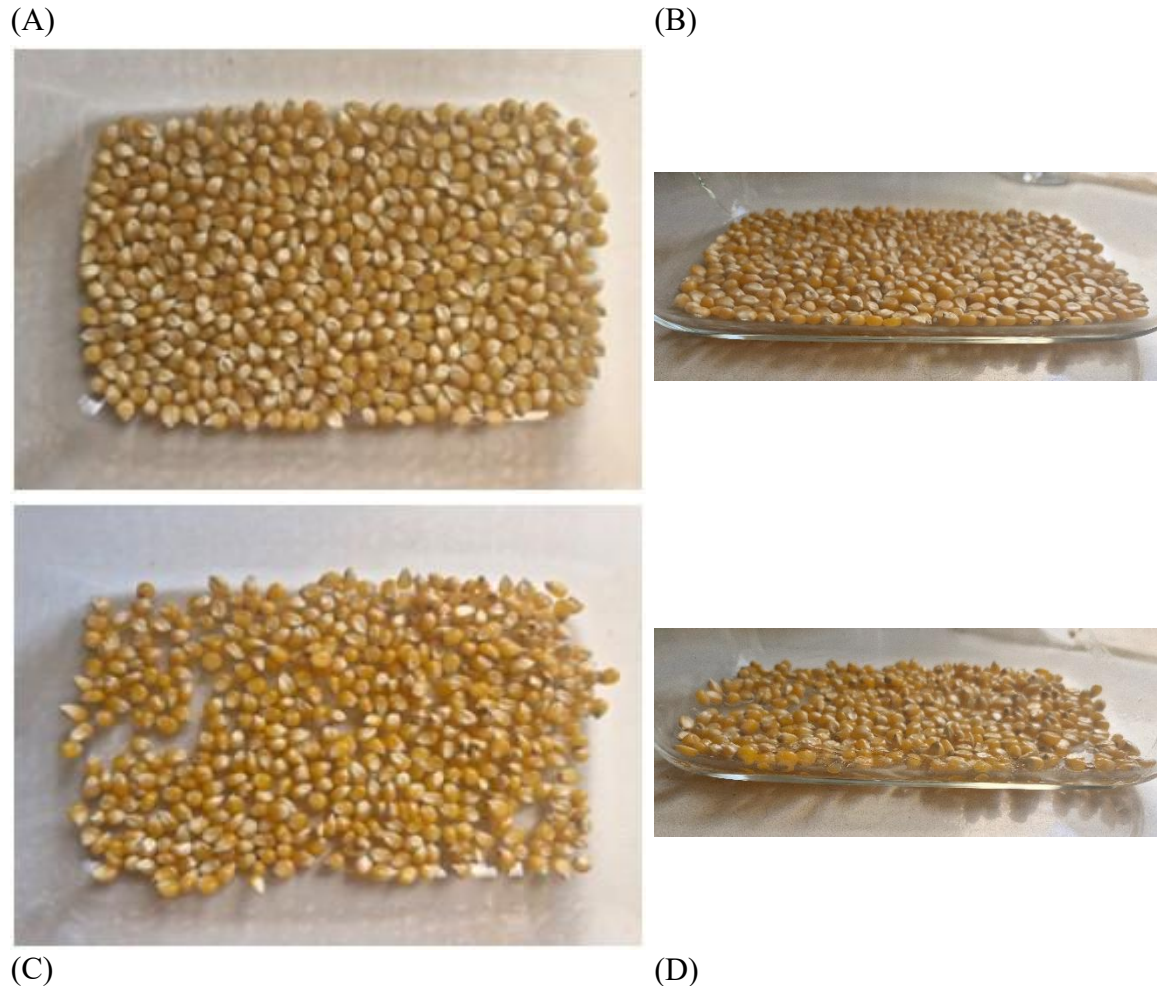
1. Introduction

The spatial organization of granular systems – collections of discrete particles – governs their macroscopic behavior and response to external forces. Understanding how particles arrange themselves, and how that arrangement changes under perturbation, is central to fields ranging from materials science to statistical physics [1, 6]. This document proposes the *Popcorn Metaphor*: a physical and conceptual framework that uses popcorn kernels to illustrate fundamental principles of order, entropy, and phase transitions in granular matter.

The metaphor exploits a simple experimental observation: dry popcorn kernels, placed in a single layer in a container, arrange themselves uniformly with minimal gaps and overlaps. When oil is added, however, the kernels cluster together, leaving voids between aggregates. This difference – uniform versus clustered – maps directly onto well-established concepts in statistical physics, including *hyperuniformity* (suppressed density fluctuations) [1] and *super-Poissonian distributions* (enhanced density fluctuations due to clustering).

The framework connects naturally to *Voronoi tessellation* analysis [2]. In a Voronoi diagram, each point (or kernel center) is assigned a cell containing all locations closer to that point than to any other. The statistical properties of these cells – their areas, shapes, and neighbor counts – encode information about the underlying spatial order. Hyperuniform systems exhibit narrow cell-area distributions (low variance), while clustered systems exhibit broad distributions with both very small cells (in dense regions) and very large cells (in voids). The entropy of the cell-area distribution provides a quantitative measure of spatial disorder.

Beyond the initial 2D arrangement, the Popcorn Metaphor extends to the popping process itself – a dramatic phase transition from rigid kernel to expanded foam [7] – and to the final jammed state of the bulk popcorn [3, 4]. This document articulates three testable claims, describes a multi-scale structural duality, and proposes an analogy to snow packing that illuminates the spectrum of jammed configurations.



Figures 1 and 2: Photographs of popcorn kernels in a glass container, illustrating the change in spatial organization due to an attractive medium. **Figure 1 (Top Row): Dry kernels**, shown in Top View (A), and Side View (B), self-organize into a uniform, hyperuniform-like distribution. **Figure 2 (Bottom Row): Oiled kernels**, shown in Top View (C), and Side View (D), aggregate into clusters separated by voids.

- The side views confirm single-layer packing for both conditions.

2. Claim A: Oil Transforms Spatial Distribution

2.1 Statement

Claim A: The addition of oil (or any short-range attractive medium) transforms the initial spatial distribution of granular particles from hyperuniform (repulsion-dominated, low variance) to clustered (attraction-dominated, high variance).

2.2 The Dry State: Repulsion and Hyperuniformity

When dry popcorn kernels are placed in a single layer within a container, they naturally arrange to fill available space with minimal gaps and no overlaps. This arrangement emerges

from *repulsive interactions*: each kernel excludes volume and cannot occupy the same space as its neighbors. The result is a quasi-uniform distribution resembling a Lloyd-relaxed point pattern or crystalline lattice.

In the language of Voronoi analysis, such a configuration is *hyperuniform* [1]: the variance in cell areas is suppressed relative to a random (Poisson) point process. Adding even one extra kernel beyond the natural packing capacity immediately creates a visible *topological defect* – a local "elevation" where kernels overlap into a second layer. In highly ordered systems, any deviation is immediately apparent; in disordered systems, additional particles blend into the noise.

2.3 The Oiled State: Attraction and Clustering

When oil is added, the system's behavior changes fundamentally. Oil introduces *short-range attractive forces* between kernels through capillary bridges and surface tension. Kernels now "stick" to one another, forming *clusters* (aggregates of high local density) separated by *voids* (regions of low density).

This clustered configuration is the opposite of hyperuniform. In statistical terms, it is *super-Poissonian*: density fluctuations are *larger* than those of a random distribution. The Voronoi cell-area distribution becomes bimodal or highly skewed, with very small cells in cluster cores and very large cells spanning voids.

2.4 Mechanism: The Shift from Intrinsic to Extrinsic Order

The transformation from hyperuniform to clustered is not merely a change in statistics – it reflects a fundamental shift in the *locus of organizational control*.

In the **dry state**, order is *intrinsic*: it emerges from direct kernel-to-kernel contact. Each kernel "knows" only its immediate neighbors through excluded-volume repulsion. The arrangement is determined by the particles themselves – their shapes, sizes, and mutual repulsion. This is *self-organization* in the strict sense.

In the **oiled state**, order is *extrinsic*: it is mediated by the surrounding medium. The oil – not the kernels – determines where particles congregate. The information about spatial arrangement has migrated *outside* the particles, into the coupling medium. This is *externally-mediated organization*.

This distinction – direct versus mediated interactions – is recognized in soft matter physics and explains *why* the addition of oil transforms spatial distribution so dramatically.

2.5 Voronoi-Based Measurement

Claim A is directly testable using Voronoi tessellation analysis:

- (1) Photograph the dry and oiled kernel arrangements from above.
- (2) Extract the (x, y) coordinates of each kernel center.
- (3) Compute the Voronoi tessellation for each configuration.
- (4) Calculate cell area statistics: mean, variance, coefficient of variation, and Shannon entropy.

(5) Compare: dry configurations should exhibit *suppressed variance* (sub-Poissonian scaling characteristic of hyperuniform systems); oiled configurations should exhibit *enhanced variance* (super-Poissonian scaling characteristic of clustered systems).

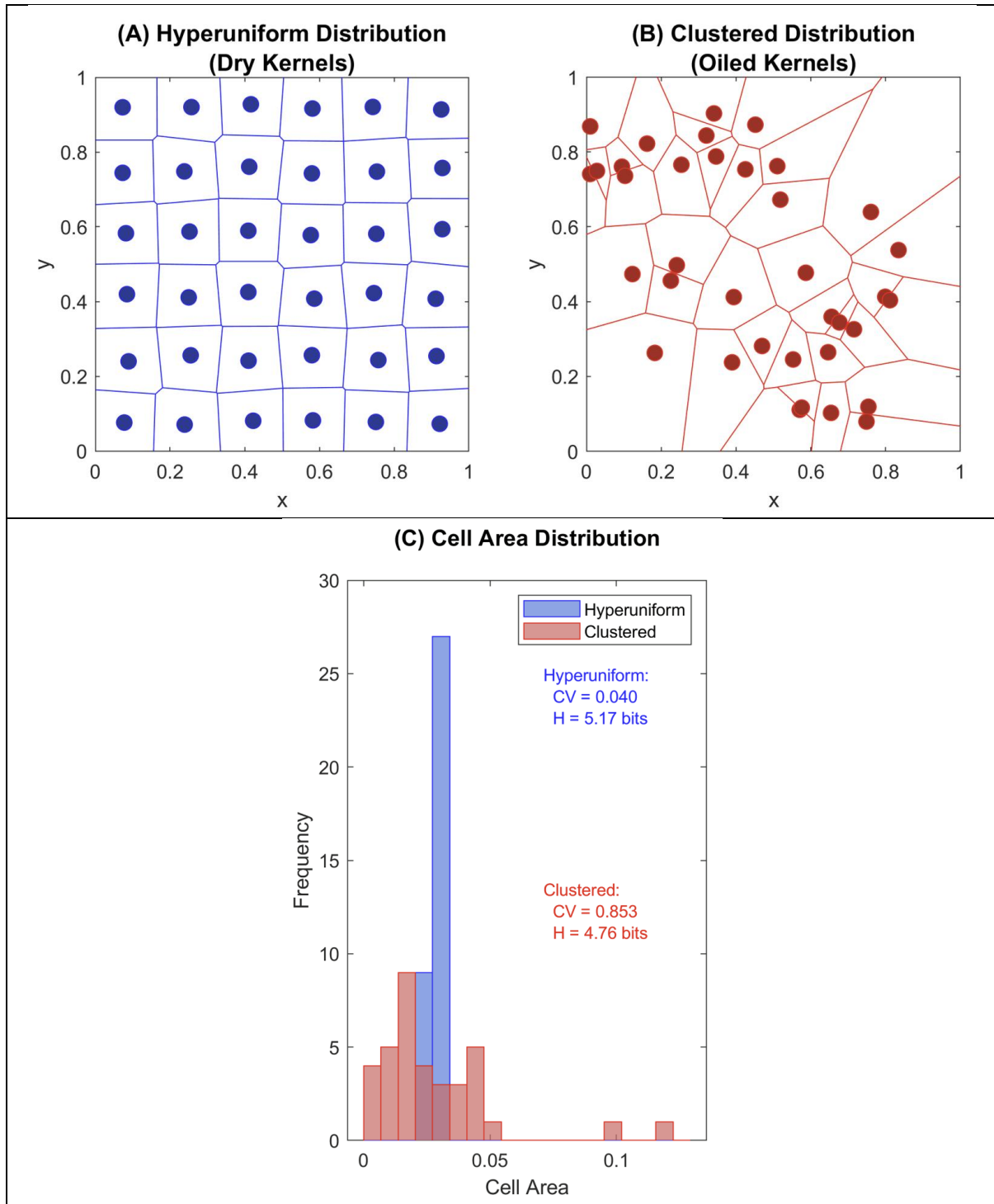


Figure 3: Voronoi tessellation comparison for hyperuniform and clustered point distributions. (A) **Hyperuniform distribution** simulating dry kernels: points self-organize into quasi-regular spacing with minimal jitter, producing Voronoi cells of similar area. (B) **Clustered distribution** simulating oiled kernels: points aggregate into distinct clusters separated by voids, producing highly variable cell areas. (C) **Histograms of cell area distributions** demonstrate the dramatic difference: hyperuniform (blue) shows tight clustering around the mean (CV = 4.02%), while clustered (red) shows broad spread (CV = 85.33%), with variance increased by a factor of 450×. Shannon entropy (5.17 bits vs 4.76 bits) reflects the uniformity of the area probability distribution, not spatial variability.

- Interpretation Note: Shannon entropy quantifies the uniformity of the probability distribution (here, cell area fractions), not spatial variability. Hyperuniform systems exhibit higher Shannon entropy because their cell areas are more uniform, creating a more uniform probability distribution. Clustered systems exhibit lower Shannon entropy despite higher spatial variability, because their cell area distribution is skewed (many small cells in clusters, few large cells in voids). The variance and coefficient of variation directly measure spatial disorder; Shannon entropy measures distributional uniformity.

3. Claim B: Initial Order Affects Temporal Dynamics

3.1 Statement

Claim B: The initial spatial order of kernels affects the temporal distribution of popping events during heating.

3.2 Popping as a Phase Transition

The **transformation** from kernel to popcorn is a *phase transition* [7]: water inside the kernel heats until it vaporizes, pressure builds, and the hull ruptures explosively. Each kernel undergoes this transition independently, but the *timing* of the transition depends on local conditions – moisture content, hull thickness, and critically, *heat delivery*.

3.3 The Temporal Signature of Order

Consider two scenarios:

Scenario 1 (Hyperuniform initial state): Dry kernels arranged uniformly in a single layer on a glass surface. Each kernel contacts the heated glass equally; heat delivery is approximately uniform. If all kernels were identical in their internal properties (moisture content, hull thickness), they would pop nearly simultaneously – a single, intense burst. In practice, intrinsic variation spreads the popping over a *narrow time window*. The frequency distribution of pops would be sharply peaked.

Scenario 2 (Clustered initial state): Oiled kernels arranged in clusters with voids. Heat delivery is non-uniform: kernels in cluster centers are insulated by neighbors; kernels at edges receive more direct heat. The result is a *broad time window* for popping. The frequency distribution would be spread, possibly multi-modal.

- *Methodological note:* Microwave heating is inherently non-uniform due to standing wave patterns in the cavity. The rotating turntable partially mitigates this, but residual spatial variation in heat delivery may contribute to temporal spread independent of initial kernel arrangement. A more controlled test of Claim B would use conductive heating (e.g., a hot plate with uniform surface temperature) to isolate the effect of initial spatial order from heating heterogeneity.

3.4 Adaptive Sampling in Time

An observer listening to popcorn popping naturally experiences this temporal signature. In the language of signal processing, one could perform *adaptive sampling*: measuring pop frequency in time windows of varying duration. At the onset and tail of popping, long windows (e.g., 10 seconds) capture sparse events; at peak activity, short windows (e.g., 0.5 seconds) are needed to resolve individual pops.

This is analogous to spatial Voronoi analysis, where window radius R varies to probe density fluctuations at different scales. Here, the window is *temporal* rather than spatial, but the principle is identical: the width of the popping distribution encodes information about the system's initial order.

3.5 Hypothesis

The temporal entropy of the popping process – quantified by the standard deviation or entropy of the inter-pop interval distribution – should correlate with the initial spatial entropy:

- Hyperuniform initial state $\rightarrow$ narrow temporal distribution (low temporal entropy)
- Clustered initial state $\rightarrow$ broad temporal distribution (high temporal entropy)

4. Claim C: Convergence to Jammed State

4.1 Statement

Claim C: Regardless of initial condition (dry or oiled), all popped systems converge toward a jammed state. The initial condition affects the path (temporal dynamics) but may also leave a trace in the quality of the final jamming.

4.2 Jamming as a Universal Endpoint

Once popping is complete, the container is filled with expanded popcorn pieces – irregular, foam-like bodies that interlock and resist further rearrangement. This is a *jammed state* [3, 4]: a disordered solid where particles are mechanically constrained by their neighbors, unable to move without coordinated rearrangement of the entire system.

The jammed state is, in a sense, a *convergent endpoint*. Whether the kernels started dry (hyperuniform) or oiled (clustered), the final bulk popcorn exhibits similar macroscopic properties: it fills the container, resists compression, and behaves as a granular solid.

4.3 The Caging Effect

A kernel that pops late in the process does not have the freedom to expand in all directions. Its neighbors – already popped and jammed – constrain its final shape and position. This is the *caging effect*: the 3D Voronoi cell available to each newly popped kernel is defined by its already-expanded neighbors.

Early poppers are "free flyers": they expand fully and settle randomly. Late poppers are "caged": they expand only into the interstices left by their neighbors. This creates a history-dependent structure where the temporal sequence of popping leaves a geometric trace in the final arrangement.

4.4 The Packing Spectrum

Jammed states are not uniform – they occupy a spectrum from *loose/fluffy* to *dense/compressed*:

- **Loose jamming:** Low packing fraction, high permeability, mechanically fragile.
- **Dense jamming (over-jammed):** High packing fraction, low permeability, mechanically rigid.

Both are jammed – mechanically stable under small perturbations – but they differ in their microscopic arrangement. Whether the initial condition (dry vs. oiled) systematically biases the final packing toward one end of this spectrum is an open empirical question amenable to Voronoi-based analysis of the final 3D configuration.

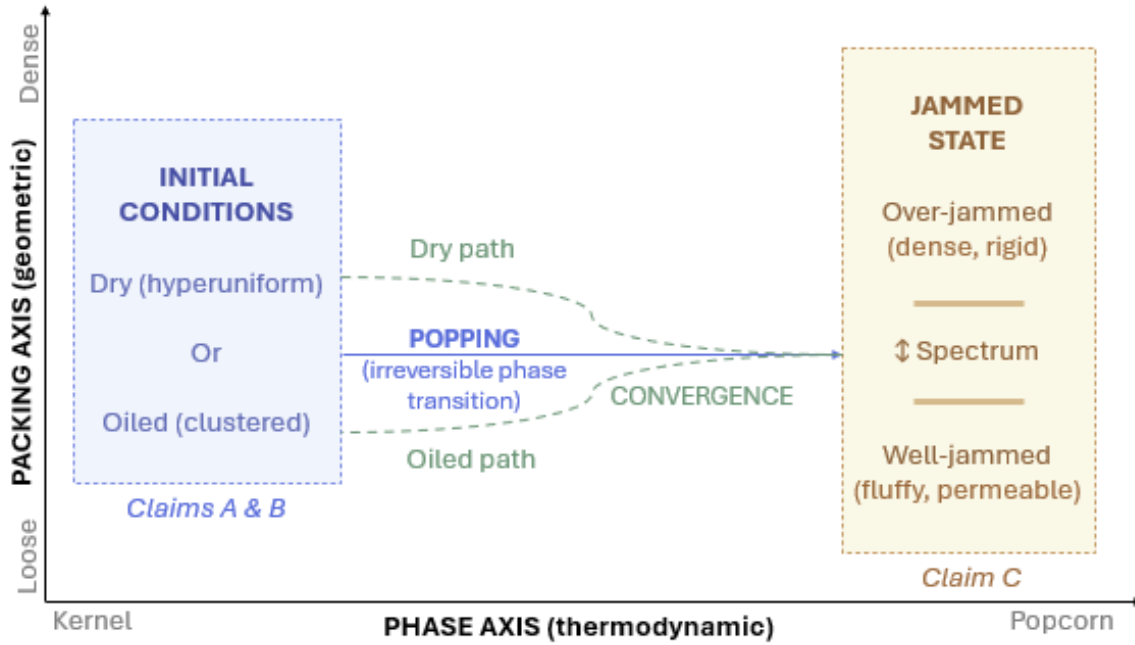


Figure 4: Two-axis framework for the Popcorn Metaphor. **Horizontal: thermodynamic phase transition** (kernel → popcorn). **Vertical: geometric packing spectrum** (loose ↔ dense). Both dry and oiled initial conditions converge to jammed states.

5. Multi-Scale Duality: Foam and Jamming

Popped popcorn exhibits a striking structural duality that depends on the scale of observation:

Table 1. Multi-scale structural duality of popped popcorn.

Scale	Structure	Property
Micro (inside each kernel)	Foam	Compressible, cellular, air-filled matrix
Macro (bulk in container)	Jammed granular system	Rigid under stress, disordered solid

At the **micro-scale**, each popcorn piece is a *foam* [5]: an expanded starch matrix riddled with air cells, formed when water vapor inflates the softened interior. Foams are compressible, lightweight, and cellular.

At the **macro-scale**, the **collection** of popcorn pieces behaves as a *jammed granular system* [3, 4]: individual units interlock, resist shear, and transmit force through contact networks. Jammed systems exhibit solid-like rigidity despite being composed of discrete, mobile units.

This *hierarchical structure* – foam inside, jammed packing outside – is not unique to popcorn. It appears in **bread** (cellular crumb, jammed loaf), metallic foams (internal pores, bulk solid), and biological tissues (cellular structure, organ-level mechanics) [5]. Similar hierarchical structures appear in geology: volcanic pumice and so-called "popcorn rocks"

exhibit foam-like internal porosity from gas expansion during cooling, while behaving as granular solids at the bulk scale. The Popcorn Metaphor thus connects to a broader class of multi-scale materials.

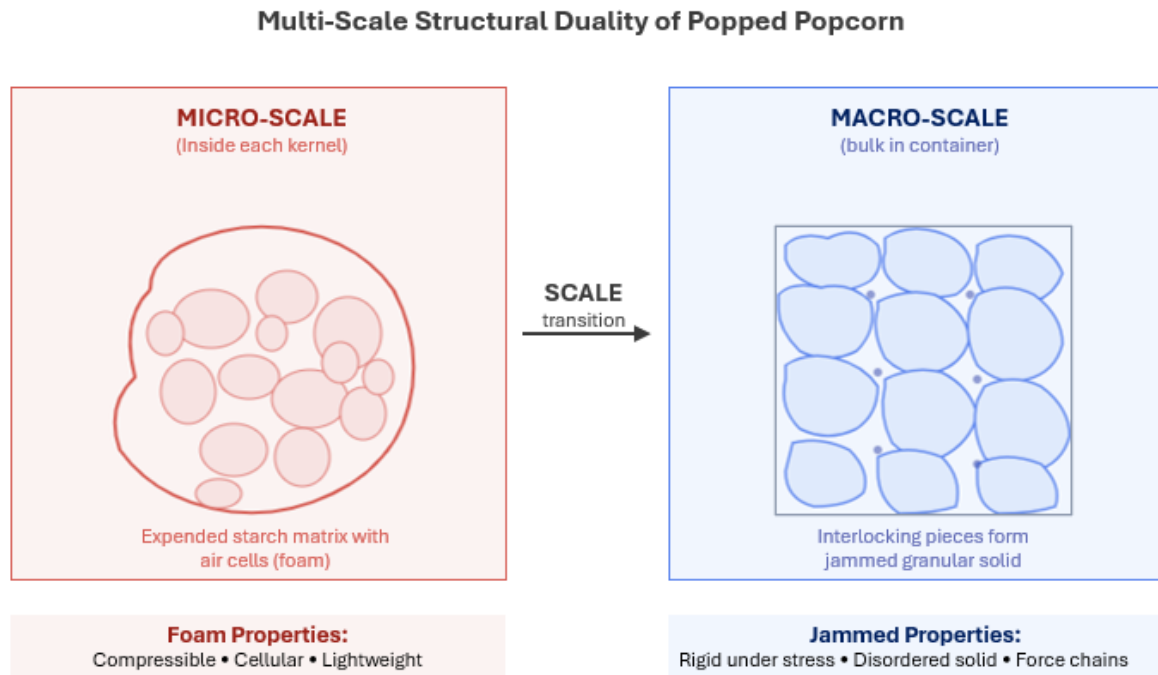


Figure 5: Multi-scale duality: each popcorn piece is foam internally (left), while the bulk collection behaves as a jammed granular system (right).

6. The Snow Analogy: A Packing Spectrum

To illustrate the spectrum of jammed states, we employ an analogy to snow. Snow exists in a single thermodynamic phase (solid water), yet it exhibits a wide range of packing configurations:

- **Powder snow:** Loose, fluffy, highly permeable, low density.
- **Packed snow:** Compressed, cohesive, moderate density.
- **Glacial ice:** Dense, crystalline, impermeable, maximum density.

These are not different phases – they are different *packing states* within the same phase. The transition from powder to packed to glacial is a matter of compression, not phase transition.

Popcorn exhibits an analogous spectrum:

- **Fluffy popcorn:** Loosely jammed, high permeability, mechanically fragile.
- **Compressed popcorn:** Densely jammed, low permeability, mechanically rigid.
- **Crushed popcorn:** Individual foam structures collapse, approaching maximum density.

The snow analogy also highlights two additional parallels:

Uniqueness: Just as every snowflake is unique (due to path-dependent crystal growth), every popped kernel is unique (due to path-dependent expansion). Both are shaped by the specific history of their formation.

Angularity: Both snowflakes and popcorn pieces are angular, non-spherical objects. Their irregular shapes affect how they pack and interlock, distinguishing them from idealized spherical particles often studied in physics.

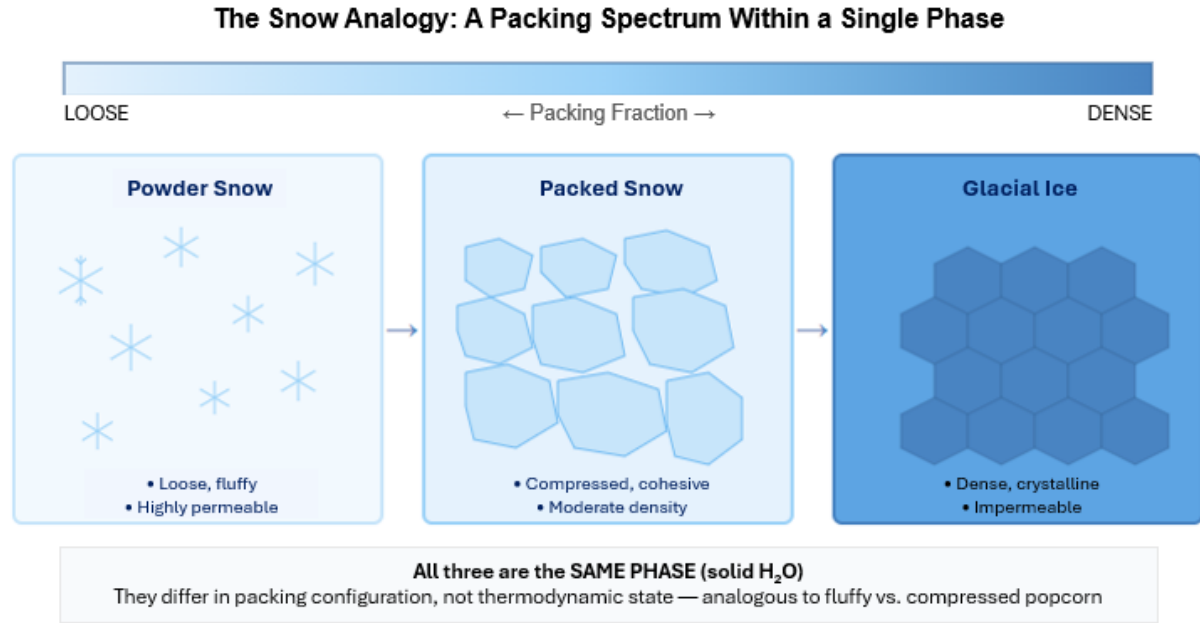


Figure 6: The snow analogy: powder snow (left), packed snow (middle), and glacial ice (right) are all solid water but occupy different positions on the packing spectrum. Popped popcorn exhibits an analogous range from fluffy to compressed within the jammed state.

7. Summary: A Two-Axis Framework

The Popcorn Metaphor can be organized along two orthogonal axes:

Phase Axis (thermodynamic): Kernel → Popcorn. This is an irreversible transformation that changes the internal structure of each unit.

Packing Axis (geometric): Loose ↔ Dense. This is a continuous spectrum within the jammed state, determined by compression and history.

Claims A and B operate along the Phase Axis, describing how initial conditions (before popping) affect spatial and temporal dynamics. Claim C operates along the Packing Axis, describing the final jammed state and its variability.

Table 2. Summary of the Popcorn Metaphor framework.

State	Dominant Force	Spatial Character	Material Analogy
Dry kernels	Repulsion	Hyperuniform	Ice (crystalline)
Oiled kernels	Attraction	Clustered (Super-Poissonian)	Water (liquid)
Popped (bulk)	Contact/friction	Jammed (spectrum)	Snow (powder → glacial)

8. Future Directions

The Popcorn Metaphor suggests several avenues for further investigation:

Quantitative Voronoi analysis: Photograph dry and oiled kernel arrangements, extract coordinates, compute Voronoi tessellations, and compare cell-area variance and entropy. This would provide a rigorous test of Claim A. MATLAB code for this analysis is provided as supplementary material.

Temporal acoustic analysis: Record the audio of popping events, extract inter-pop intervals, and compute temporal entropy. Correlate with initial spatial order (using uniform conductive heating) to test Claim B.

Packing fraction measurements: Measure the final packing fraction of popped popcorn as a function of initial condition (dry vs. oiled) to test whether initial state biases final jamming quality.

Kernel-level information encoding: Investigate whether the internal structure of each kernel (moisture content, hull thickness, starch distribution) can be measured and used to predict the shape and size of the resulting popcorn. This would treat each kernel as an *information container* – a discrete unit whose internal state encodes, and whose final form reveals, the information present at the moment of transformation. The development of this concept is left for future work.

9. Conclusion

The Popcorn Metaphor provides a tangible, experimentally accessible framework for understanding order, entropy, and phase transitions in granular systems. By manipulating a single variable – the presence or absence of a coupling medium – one can shift a system from hyperuniform to clustered, from intrinsically to extrinsically organized, from low to high spatial entropy.

The framework connects naturally to Voronoi tessellation analysis, offering quantitative predictions for variance and entropy that can be tested experimentally. It extends beyond spatial organization to temporal dynamics (how the system transitions) and final states (how it jams), providing a unified view of granular behavior across multiple stages.

The multi-scale duality of popped popcorn – foam inside, jammed outside – illustrates how structure and behavior depend on the scale of observation, connecting this simple kitchen experiment to broader themes in materials science and soft matter physics.

Finally, the snow analogy reminds us that phase and packing are orthogonal dimensions: a material can remain in a single phase while traversing a wide spectrum of packing configurations. Understanding this distinction is essential for interpreting both the Popcorn Metaphor and the physical systems it represents.

Acknowledgments

This conceptual framework emerged through extended dialogue and collaborative refinement. The author thanks all interlocutors who contributed to sharpening these ideas.

References

- [1] Torquato, S., & Stillinger, F. H. (2003). Local density fluctuations, hyperuniformity, and order metrics. *Physical Review E*, 68(4), 041113.
- [2] Aurenhammer, F. (1991). Voronoi diagrams – a survey of a fundamental geometric data structure. *ACM Computing Surveys*, 23(3), 345–405.
- [3] Liu, A. J., & Nagel, S. R. (1998). Jamming is not just cool any more. *Nature*, 396(6706), 21–22.
- [4] O'Hern, C. S., Silbert, L. E., Liu, A. J., & Nagel, S. R. (2003). Jamming at zero temperature and zero applied stress: The epitome of disorder. *Physical Review E*, 68(1), 011306.
- [5] Gibson, L. J., & Ashby, M. F. (1997). *Cellular Solids: Structure and Properties* (2nd ed.). Cambridge University Press.
- [6] Jaeger, H. M., Nagel, S. R., & Behringer, R. P. (1996). Granular solids, liquids, and gases. *Reviews of Modern Physics*, 68(4), 1259–1273.
- [7] Viot, E., & Ponomarenko, A. (2015). Popcorn: critical temperature, jump and sound. *Journal of the Royal Society Interface*, 12(104), 20141247.

Supplementary Material

The following supplementary materials are available:

- *Popcorn_Voronoi_Analysis.m*: MATLAB code for generating hyperuniform and clustered point distributions, computing Voronoi tessellations, and calculating cell area statistics.
- *Popcorn_Voronoi_Statistics.csv*: Summary statistics table from the MATLAB analysis including mean area, variance, coefficient of variation, and Shannon entropy for both distributions.
- *Figures 3–6* in PNG format for high-resolution reproduction.